

Hodge Laplacian $L_k = \underbrace{\partial_{k+1} \partial_{k+1}^*}_{L_k^{\text{row}}} + \underbrace{\partial_k^* \partial_k}_{L_k^{\text{column}}}$
 $\hookrightarrow \ker(L_k) \cong H_k(M; \mathbb{R})$
available local columns of the space

Betti's are Laplacian nullities

$\hookrightarrow k=0, L_0 = \partial_1 \partial_1^*$ is graph Lapl for diffusion
 $\dim \ker L_0 = \beta_0$

$\hookrightarrow$ Diff geom and No sum operation
non zero spectrum nulling

Persistent Lapl $\Delta_k^{\text{p.c.}}$ has nullity equal to persistent Betti
 smallest nonzero real tracks given

PCA, classical MDS on centered Gram matrix is

large bandwidth limit of diff. kernel $K_{ij} = e^{-\|x_i - x_j\|^2 / \epsilon}$
 $\sim 1 - \|x_i - x_j\|^2 / \epsilon$

by orthonormal principal axes